

Natural Sciences

CLASS TEXT & STUDY GUIDE

Mariechen Vermeulen, Grace Elliott, Liesl Sterrenberg,
Retha Louw, Silvana Scarola & Norman Davies

GRADE

8

CAPS

3-in-1



THE
ANSWER
SERIES *Your Key to Exam Success*



Grade 8 **Natural Sciences** 3-in-1 CAPS

CLASS TEXT & STUDY GUIDE

This jam-packed, full-colour study guide naturally transitions learners from basic scientific concepts to the skilled application of knowledge required in the FET phase. It follows closely in the footsteps of the best-selling Grade 9 Natural Sciences book.

It includes:

- Comprehensive Skills Section
- Detailed Answers
- Organised, easy-to-follow Notes
- Full-colour hard copy and eBook
- Questions per Topic

Key Features:

- Skills section:
 - step-by-step explanation of the scientific method
 - worked example of a scientific investigation question
 - illustrated summary of representing data (tables/graphs/diagrams)
 - curriculum-aligned, comprehensive yet compact NOTES per Topic
- clear, self-explanatory VISUAL SUMMARIES and ILLUSTRATIONS
- beautiful PHOTOGRAPHS and IMAGES to enrich the text and stimulate interest
- step-by-step, ILLUSTRATED PRACTICALS with results and conclusions
- extensive QUESTIONS and ANSWERS per Topic
- detailed MEMOS with handy hints in a separate booklet
- teacher-TIPS throughout the text
- ENRICHMENT to stimulate critical thinking and connect curricular content to the real world

This long-awaited newcomer provides an easy-to-follow, reliable introduction to the diverse topics covered in this compulsory and challenging subject. The learner-friendly style encourages independent learning, sparks curiosity for the Sciences and grows confidence in mastering the content.

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THIS CLASS TEXT & STUDY GUIDE INCLUDES

1

Notes

- Life and Living
- Matter and Materials
- Energy and Change
- Planet Earth and Beyond

2

Questions per Module

3

Detailed Answers

separate
ANSWER
BOOKLET

eBook
available



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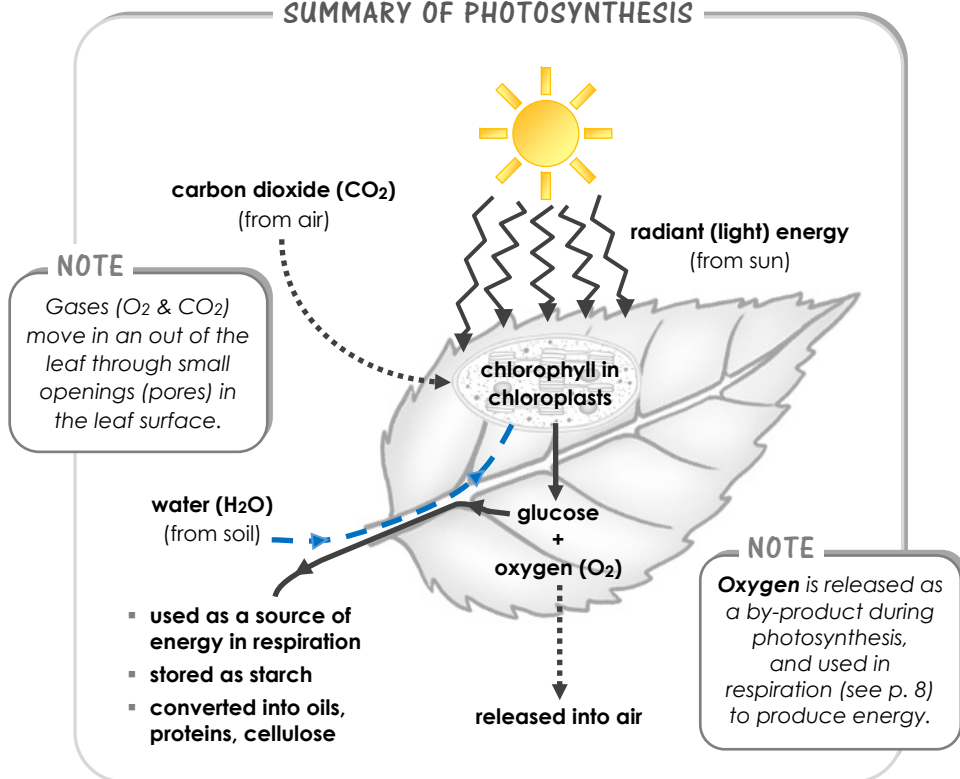
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SUMMARY OF PHOTOSYNTHESIS



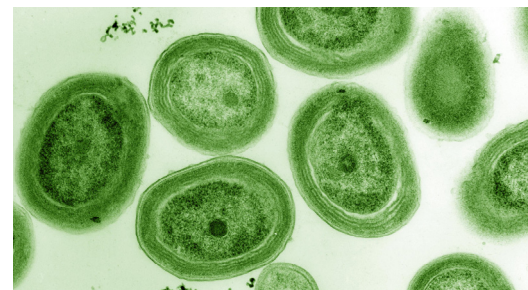
DID YOU KNOW?

Plants are not the only organisms that contain chlorophyll. Algae, certain microorganisms and even some animals can also photosynthesise to produce their own food.

Elysia chlorotica is a green sea slug that absorbs chloroplasts from its plant food to photosynthesise



Smithsonian Environmental Research Center, CC BY 2.0
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Cyanobacteria that can photosynthesise



Green algae under the microscope

NOTE

When an organism can perform photosynthesis we say it is **photosynthetic**.



STORAGE AND USE OF GLUCOSE

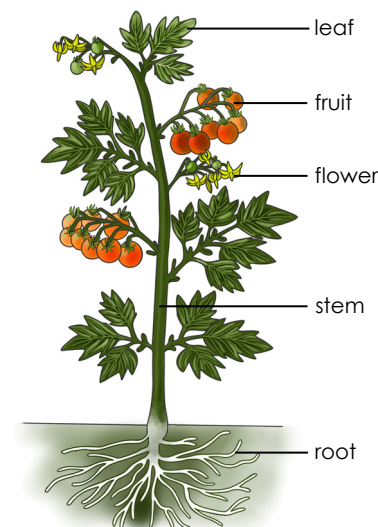
- Glucose is a small, **soluble** molecule that can dissolve in water.
- Dissolved glucose can be easily transported from the leaves to the roots, stems, fruits and flowers.
- If glucose is not used immediately, it must be stored in the **plant organs** (stems, roots, etc.) in an **insoluble** form.



soluble: dissolves in water

insoluble: cannot dissolve in water

- Starch** is a larger, stable, **insoluble** molecule made up of glucose molecules. It is the storage form of glucose in plants.



Organs of a flowering plant

LEVELS OF ECOLOGY

Living **organisms** and the interactions between them and their non-living environment are organised into four levels of **ecological interactions**. These levels arranged from small to large are: **populations**, **communities**, **ecosystems** and the **biosphere**.



Organism

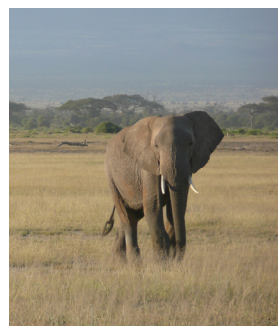
An organism is an **individual** (plant or animal) of a particular **species** living in its natural **habitat**.



Meerkat



Sea anemone



Elephant



Acacia tree

Different organisms in their natural habitats

NOTE

Habitats can be **terrestrial** (on land) or **aquatic** (in water).



species: organisms that look similar, can interbreed, and produce fertile offspring



Population

A population is a **group of individuals** of the **same species** living in the **same area** at the **same time** that can **interbreed** with each other.



Colony of ants



Herd of buffalo

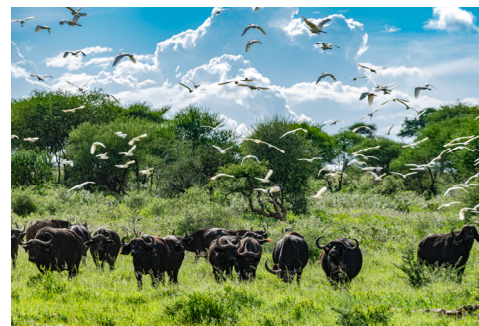


Pine tree plantation

Different populations in their habitats

Community

A community consists of **different populations** that occur in the **same area** at the **same time** and **interact** with each other.



A community in Tarangire National Park, Tanzania



A community in a coral reef

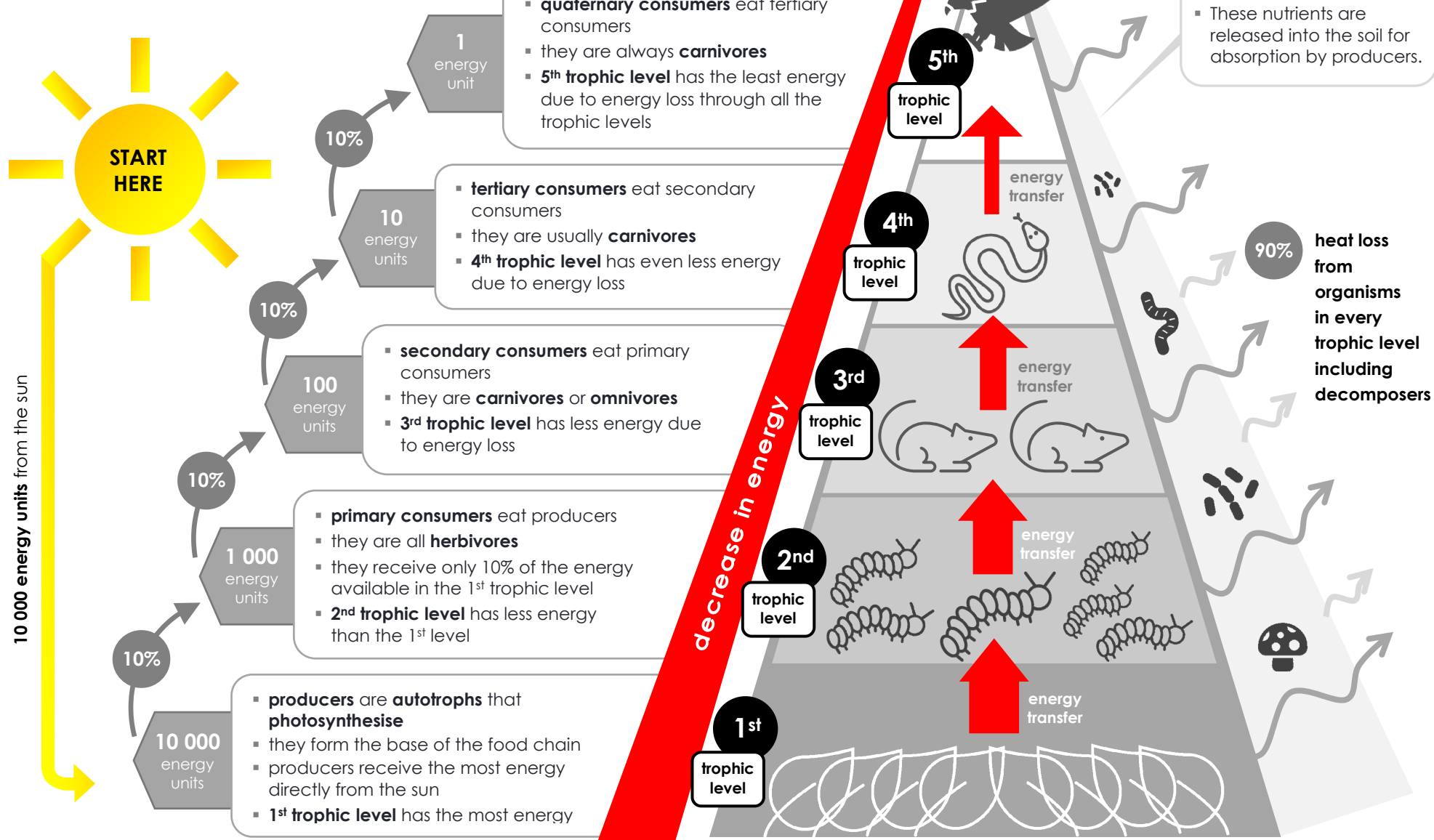
Nina R via Flickr
https://www.flickr.com/photos/150102727@N06/51468158824/

SUMMARY OF ENERGY PYRAMID



NOTE

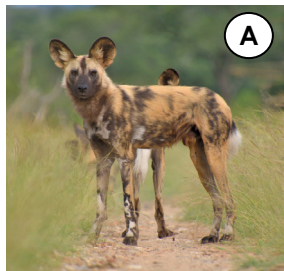
As the amount of available energy per trophic level decreases, so does the number of organisms present on each trophic level.



- 17.1 Why would a predator need camouflage?
- 17.2 List all the structural adaptations provided in the diagram that enable the cheetah to reach high speeds.
- 17.3 Sharks are also excellent predators of the ocean ecosystem. Provide at least THREE similarities between cheetahs and sharks.
- 17.4 Sharks are nocturnal predators. How does the colouring of a shark support this behavioural adaptation?

Question 18

The images **A** to **F** below show different types of adaptations (structural/functional/behavioural) in organisms.



African wild dogs hunt in packs



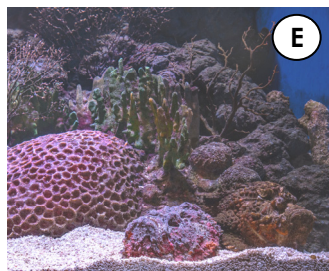
An octopus can change its colour and texture



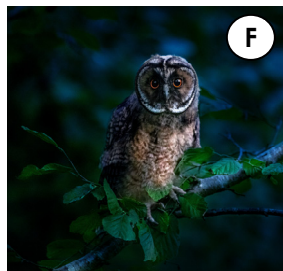
A bright blue dart frog



The owl butterfly's markings look like eyes



Two predatory stone fish (bottom right corner)



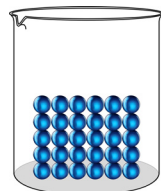
Owls hunt at night

- 18.1 Image **A**:
- 18.1.1 What type of carnivore is an African wild dog?
- 18.1.2 What is the benefit of African wild dogs hunting in packs?
- 18.1.3 What type of adaptation is illustrated by this hunting strategy?
- 18.2 Image **B**:
- 18.2.1 What type of adaptation is represented here?
- 18.2.2 Another example of this type of adaptation is warning colouring. Give the LETTER (**A** to **F**) of an organism that displays this feature.
- 18.2.3 What functional adaptation does the warning colouring of an animal indicate?
- 18.3 Image **D**:
- 18.3.1 What special type of structural adaptation is exhibited here?
- 18.3.2 Explain the benefit of this adaptation for the owl butterfly.
- 18.4 Image **E**:
- 18.4.1 Why do you think these fish are called stone fish?
- 18.4.2 What special type of structural adaptation is exhibited here?
- 18.4.3 Stone fish are predators. Explain how they would benefit from the adaptation mentioned in QUESTION 18.4.2.
- 18.5 Image **F**:
- 18.5.1 What special type of behavioural adaptation is exhibited here?
- 18.5.2 Provide TWO reasons why this type of adaptation is beneficial to animals.

THE STATES OF MATTER

Solids

Solids have a **fixed shape** and are **not compressible**, i.e. they cannot be packed more tightly due to the very small spaces between their particles.

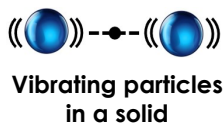


NOTE

A **compressible material**, e.g. a gas (see p. 88), can be compacted or packed into a smaller space/volume, but a solid is not compressible.

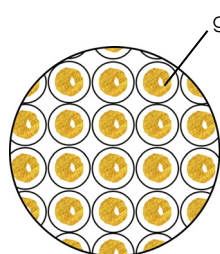
Particles in a solid:

- 1 are **closely packed** and arranged in a **regular, repeating pattern**
- 2 have **very small spaces** between them
- 3 do not move freely – only **vibrate** in fixed positions
- 4 are held together with **strong forces of attraction**
- 5 have very **little kinetic energy** (energy of motion)

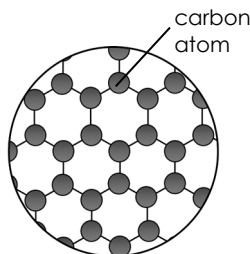


NOTE

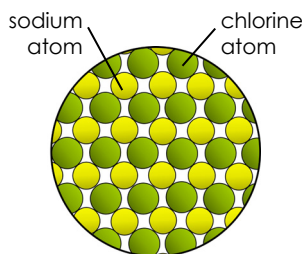
Kinetic energy is the energy of motion (movement), and depends on the speed of motion. The faster an object moves, the more kinetic energy it has. As the particles in **solids** cannot move freely and **only vibrate**, they have **very little kinetic energy**.



The element gold (Au) has a regular pattern of closely-packed gold atoms



The atoms of some solids, like carbon (C), are arranged into a repeating, crystalline pattern

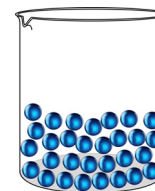


The compound table salt (NaCl) has closely-packed sodium and chloride atoms

Examples of regular patterns in solids

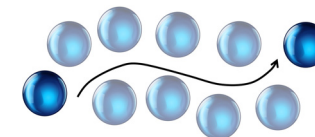
Liquids

Liquids **flow** and are **not compressible**. They **take on the shape** of the container they occupy. Examples include water, liquid mercury and sugar water.



Particles in a liquid:

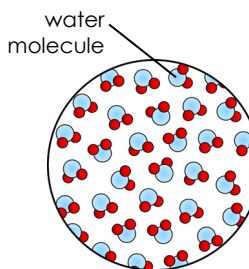
- 1 are **randomly** and **loosely arranged**, but still **quite close together**
- 2 have **small spaces** between them
- 3 move fast by **flowing** and **sliding past each other**
- 4 are held together by **weaker forces of attraction**
- 5 have **more kinetic energy** than solids



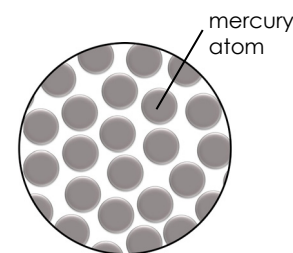
Sliding particle in a liquid

NOTE

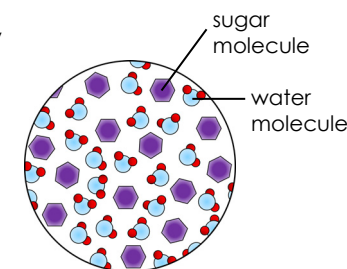
The kinetic energy of the particles in a liquid will depend on the speed at which they move around. When a liquid is **heated**, the particles **move faster** and **gain kinetic energy**.



The compound water (H₂O) is a liquid



The element mercury (Hg) is a liquid at room temperature



The mixture sugar water is a liquid

Examples of random patterns in liquids



NOTE

A liquid takes the shape of any container it occupies, because its particles can slide past each other to fill the spaces left by other particles.

○ Signs that a **chemical change has taken place** include:

- when a substance **changes colour**
- when a **gas is produced** (this will appear as bubbles in a liquid)
- when a **new smell** can be **detected**
- when a **precipitate** has **formed**



precipitate:
a solid substance that forms in a liquid solution

UNIT

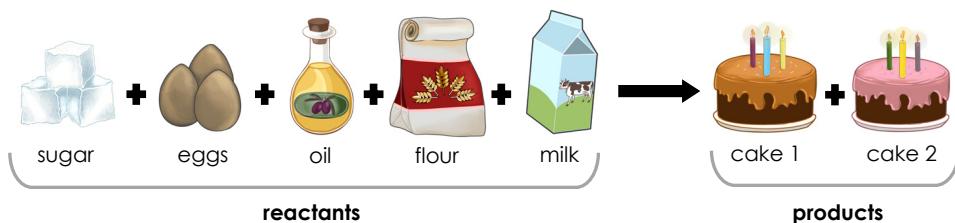
2

REACTANTS & PRODUCTS

○ Chemical changes occur during **chemical reactions**.



chemical reaction: a process during which chemical substances (reactants) react with each other to produce new chemical substances (products)



A simple chemical reaction

- The ingredients to bake the cakes represent different chemical **substances that react with each other**. These **reactants** are **present at the start** of a chemical reaction.
- The two cakes represent **substances that are produced** due to the reaction between the ingredients. These **products** are **formed at the end** of a chemical reaction.

○ **Reactants** therefore combine and change chemically during a chemical reaction to **form** new, different substances called **products**.



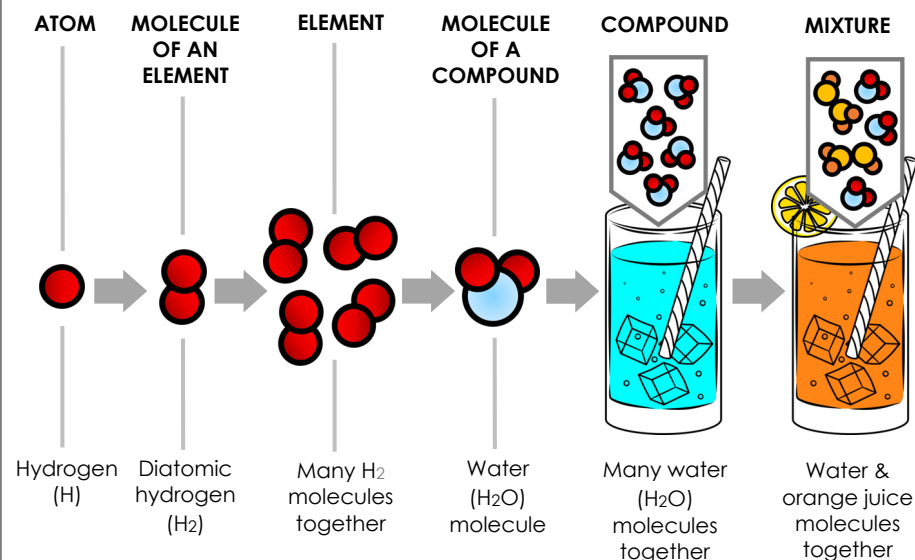
reactants: substances that are present at the start of a chemical reaction

products: substances present at the end of a chemical reaction

REMEMBER

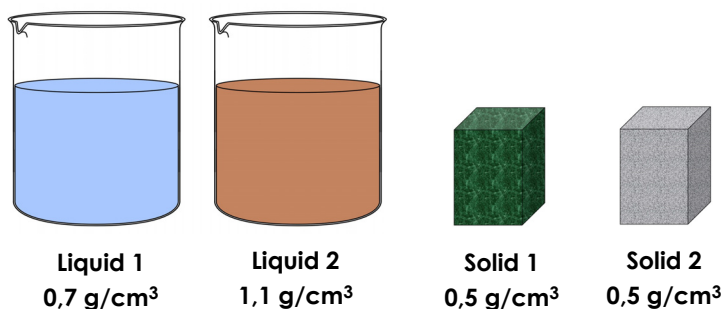
Use the diagram below and recap the important terms from TOPIC 1:

- atom
- diatomic molecule
- element
- compound
- mixture



Question 16

Peter has two liquids (**1** and **2**) as well as two solids (**1** and **2**) in his possession. The density of each liquid and solid is shown below.



For each of the following statements, indicate whether it is *true* or *false*. In each case give a reason for your answer.

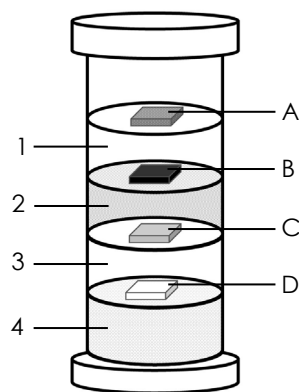
- 16.1 Solid **1** will float on both Liquid **1** and Liquid **2**.
- 16.2 Liquid **1** will float on Liquid **2** if they are not mixed.
- 16.3 Solid **2** will float on Liquid **2**, but sink in Liquid **1**.
- 16.4 Solid **1** will float deeper in Liquid **1** than in Liquid **2**.

Question 17

The apparatus below shows four liquids (**1 – 4**) in a cylinder. Four solids (**A** to **D**) can be seen floating on the liquids.

The table below shows the relative densities of the liquids (**1** to **4**) and solids (**A** to **D**) in the cylinder alongside.

Liquid density (g/cm ³)		Solid density(g/cm ³)	
mercury	13,6	ice	0,91
alcohol	0,79	silver	10,5
glycerine	1,26	wood (ebony)	1,2
water	1,0	polystyrene	0,64

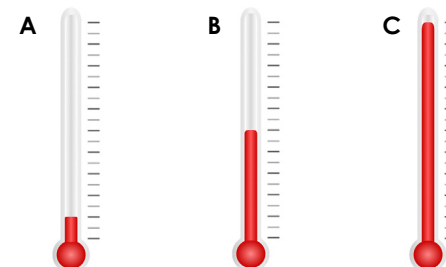


17.1 Identify the liquids **1** to **4**.

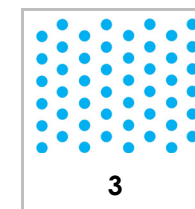
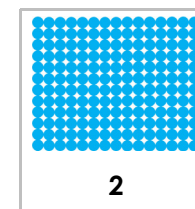
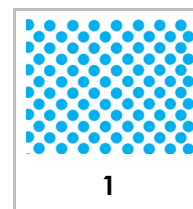
17.2 Identify the solids **A** to **D**.

Question 18

Pictures **A**, **B** and **C** show three thermometers.



- 18.1 What toxic element is used in some thermometers? Provide its NAME and chemical SYMBOL.
- 18.2 Which phenomenon in matter is used in a thermometer?
- 18.3 The three representations below show the liquid particles in the three thermometers provided. Which number (**1** to **3**) best represents the liquid particles inside each of the thermometers (**A** to **C**)?



- 18.4 For each of the following statements, indicate whether it is *true* or *false*:
 - 18.4.1 When a material contracts, its mass decreases.
 - 18.4.2 The volume of a material increases when it is heated.
 - 18.4.3 When a material expands, its particles get bigger.
 - 18.4.4 When a material contracts, its particles move closer to one another.
 - 18.4.5 The density decreases when a material expands.

- Resistors connected in an electric circuit influence the **amount of electric current** (i.e. its **strength**) in a circuit.



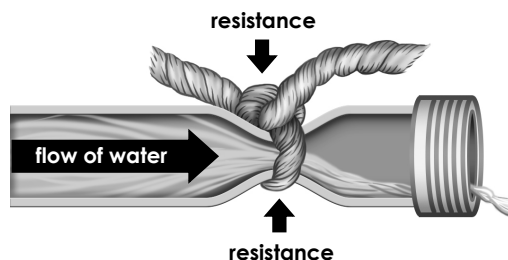
resistor: a poor conductor that resists/opposes the flow of an electric current

- The **higher** the **resistance** in a circuit, the **weaker** the **current** that flows through the circuit.
- The **lower** the **resistance** in a circuit, the **stronger** the **current** that flows through the circuit.

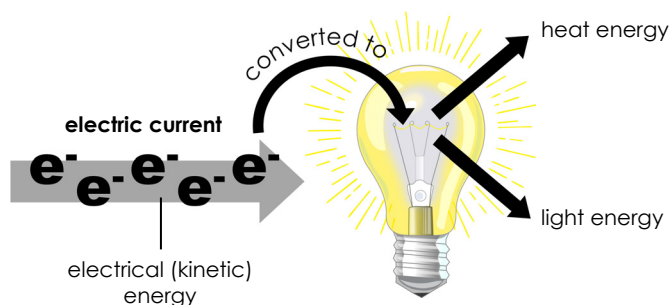


NOTE

When you squeeze a hose pipe, there is a weaker stream of water flowing out. In the same way resistors decrease the flow of an electric current.



- Some **resistors** (light bulb filaments, heating wires, elements in kettles/heaters/geysers/stoves etc.) **can heat up** to provide useful output energy.
- The electrical energy is **converted into** other types of energy in the process, e.g. **heat** or **light** energy.

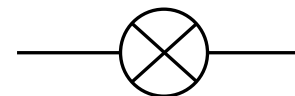


Energy conversion in a light bulb

LIGHT BULBS



SYMBOL



Light bulbs contain a resistance wire called a **filament**.

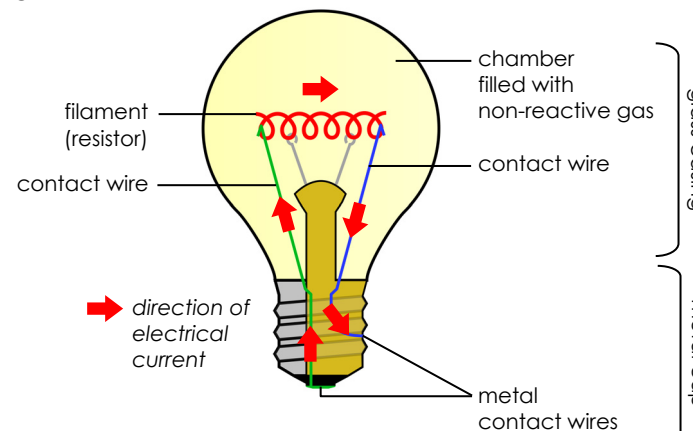
- The filament acts as a **resistor** and **converts electrical energy to heat and light energy**.
- Light bulbs are used in circuits to **determine** if **current** is **flowing**.
 - If the light bulb is **not glowing**, it indicates that **no current is flowing** through the circuit due to an **open circuit** or **wrong connections**.
 - If the light bulb is **glowing**, it indicates that the **circuit is closed** and **current is flowing** through the circuit to the light bulb.



incandescent: to emit light as a result of heating

Structure of a light bulb

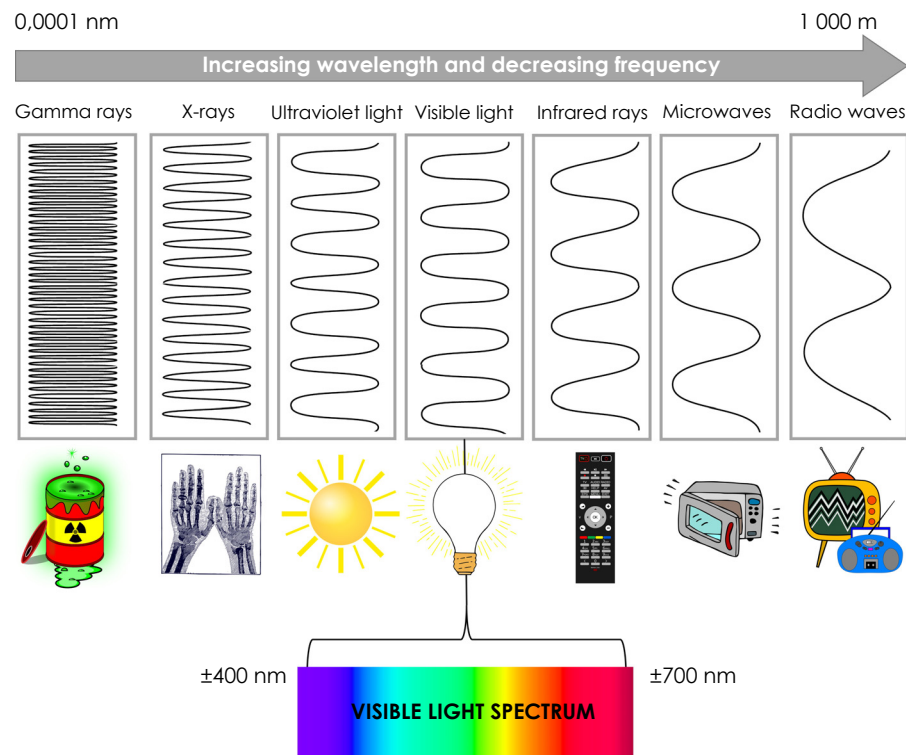
- A standard, **incandescent** light bulb consists of:
 - a **glass casing** filled with a non-reactive gas
 - a **metal cap** that encloses the base of the light bulb



Structure of an incandescent light bulb

THE ELECTROMAGNETIC SPECTRUM

- different types of electromagnetic radiation -



NOTE

nm is the symbol for **nanometre** (one billionth of a metre or 10^{-9} m)



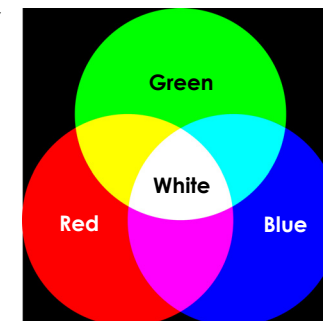
- A small section of the electromagnetic spectrum is composed of wavelengths and frequencies that may be **detected by the human eye**.
- This section is known as the **visible light spectrum** and lies between wavelengths 400 nm and 700 nm.
- **Seven colours** can be distinguished in the visible light spectrum:

violet indigo blue green yellow orange red

- **All the colours** of the visible light spectrum combine to form **white light**.

NOTE

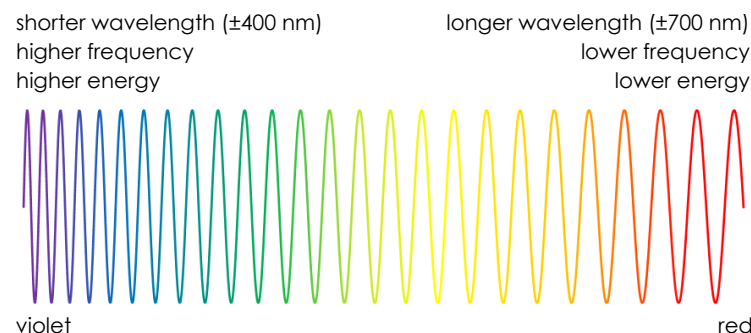
- **Red, green and blue** are the **primary colours of light**.
- **Other colours** like pink and brown are **mixtures of the seven colours of visible light**.
- **White** is not the absence of colour, but **all the colours** in the visible light spectrum **combined**.



- Each colour of the visible light spectrum has its own **wavelength, frequency and energy**:
 - **violet light** has the **shortest wavelength, highest frequency and highest energy**
 - **red light** has the **longest wavelength, lowest frequency and lowest energy**

NOTE

The **colour** of light we see **depends on the wavelength and frequency** of the light rays that enter the eye.



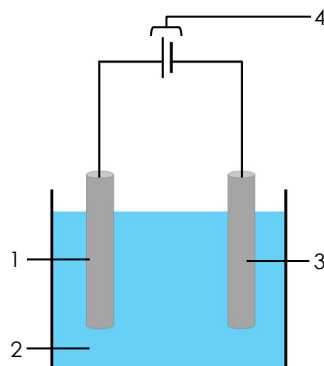
The wavelengths, frequencies and energy of the different colours in the visible light spectrum

DScComm, CC BY-SA 4.0
<https://creativecommons.org/licenses/by-sa/4.0/> via Wikimedia Commons

Question 10

In the accompanying experiment, an electric current is passed through a solution of copper chloride (CuCl_2). Components **1** and **3** represent electrodes.

Refer to MODULE 2, TOPIC 1 p. 82 to recap electrolysis in detail.



- 10.1 Which property/effect of an electric current is illustrated by this experiment?
- 10.2 Identify the term used for the process illustrated above.
- 10.3 What are the electrodes composed of?
- 10.4 Which NUMBER represents the:

10.4.1 anode	10.4.2 cathode	10.4.3 electrolyte
--------------	----------------	--------------------
- 10.5 What is the purpose of number **4** in the experiment?
- 10.6 During the experiment, gas bubbles form at one of the electrodes.
 - 10.6.1 Name the electrode at which gas bubbles will form.
 - 10.6.2 Which gas forms the bubbles at this electrode?
 - 10.6.3 List ONE other observation that will confirm the gas identified in QUESTION 10.6.2.
 - 10.6.4 Briefly explain what is observed at the other electrode.
- 10.7 Why is it important that the two electrodes do not touch each other while the electric current is flowing?
- 10.8 This process is used during electroplating.
 - 10.8.1 List TWO everyday items that may be electroplated.
 - 10.8.2 Give THREE reasons for the electroplating of items.

Question 11

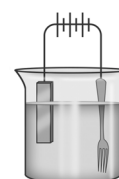
There are many energy conversions that occur in an electrical system. Identify the energy conversion that occurs within each of the systems (**A** to **F**) shown below.



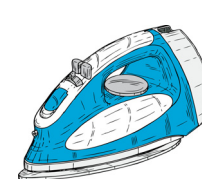
A



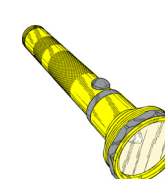
B



C



D



E



F

TOPIC 3

SERIES & PARALLEL CIRCUITS

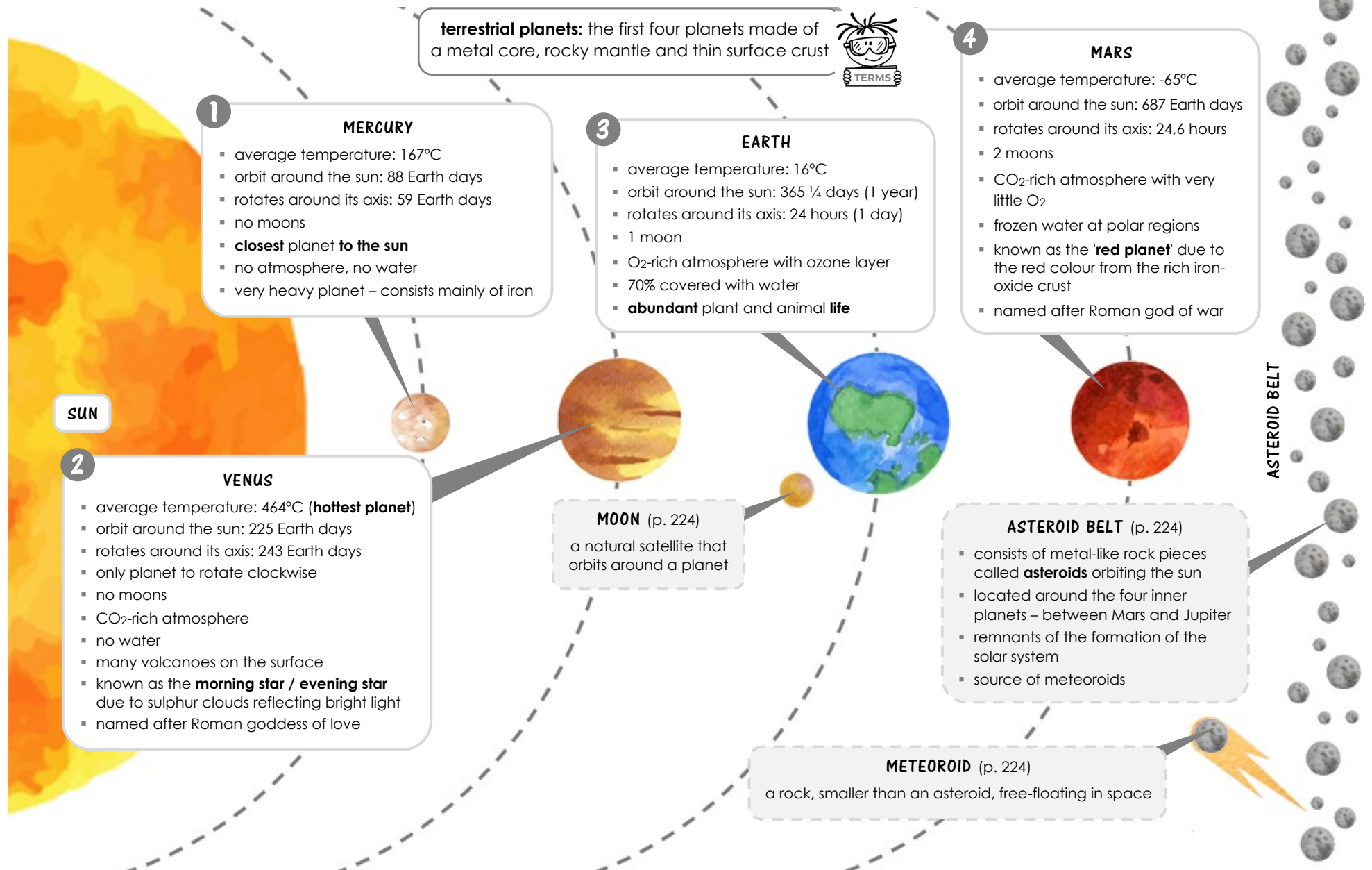
Question 1

Give ONE word or term for each of the following statements:

- 1.1 A circuit in which current can only flow along one pathway.
- 1.2 A circuit in which current flows along more than one pathway.
- 1.3 Circuit components that oppose the flow of charge and decrease the strength of a current.
- 1.4 The metal commonly used in electric cables to ensure fast and easy supply of electricity.
- 1.5 The metal alloy commonly used in the elements of electrical appliances.
- 1.6 Devices that convert electrical energy into useful forms of energy.
- 1.7 Lights that are an energy-efficient alternative to the standard incandescent light bulb.
- 1.8 The type of effect produced by an electric current and used in the functioning of an electric motor.

THE SMALLER, INNER TERRESTRIAL PLANETS

The **four inner planets** are relatively small and have rocky surfaces. They are called **terrestrial planets**. They all have a metal core, rocky mantle (middle layer) and thin surface crust. The **asteroid belt** separates the four inner planets from the four outer planets.



CALCULATIONS OF DISTANCE IN SPACE

Use the information in the tables provided.

Our solar system has a diameter of about 13 light hours.

How many kilometres is 13 light hours?

Distance in kilometres
 $= 13 \times 1\,000\,000\,000$
 $= 13\,000\,000\,000$ (13 billion km)

NOTE

1 hour = 1 billion km

How many light minutes is 13 light hours?

Distance in light minutes
 $= 13 \times 60$
 $= 780$ light minutes

REMEMBER

1 light hour = 60 light minutes

How many light seconds is 13 light hours?

Distance in light seconds
 $= 780 \times 60$
 $= 46\,800$ light seconds

REMEMBER

1 light minute = 60 light seconds

LOOKING BACK IN TIME

ENRICHMENT

When we measure distance in terms of light, we are really expressing how long it takes light from the object to reach us.

Light takes 8 minutes to reach the earth from the sun. This means that if the sun were 'switched off' like a lamp, darkness would only be experienced 8 minutes later.

Light that is emitted from an object that is one light year away from Earth, takes one calendar year to reach the earth.

When viewing Alpha Centauri in the Southern Cross, we are really seeing the star as it would have appeared 4,2 years ago. It took light 4,2 years (light years) to reach the earth from Alpha Centauri.

When scientists view space and its celestial objects through telescopes, the images they capture are technically in the past – i.e. they are 'looking back in time'. It is like looking at a photograph of an old man when he was a little boy. Photos of space show what celestial objects looked like when light was first emitted from them.



This image shows the Helix Nebula (690 light years away). Light emitted by this giant cloud of dust and gas travelled 690 years before reaching Earth. So, this image shows what it looked like 690 years ago.



The Andromeda galaxy is 2,5 million light years from Earth



The Fairy of Eagle Nebula is 9,5 light years from Earth